

Social Innovation Hackathon

e- Setch Bin

A Smart Solution to Rural Waste Management

**BVRIT HYDERABAD College of
Engineering for Women**



Problem



Rural India is dealing with a growing waste issue, mainly from increasing domestic waste



The Ministry of Drinking Water and Sanitation reports an estimated 0.3 to 0.4 million metric tons of solid waste generated daily in rural households,

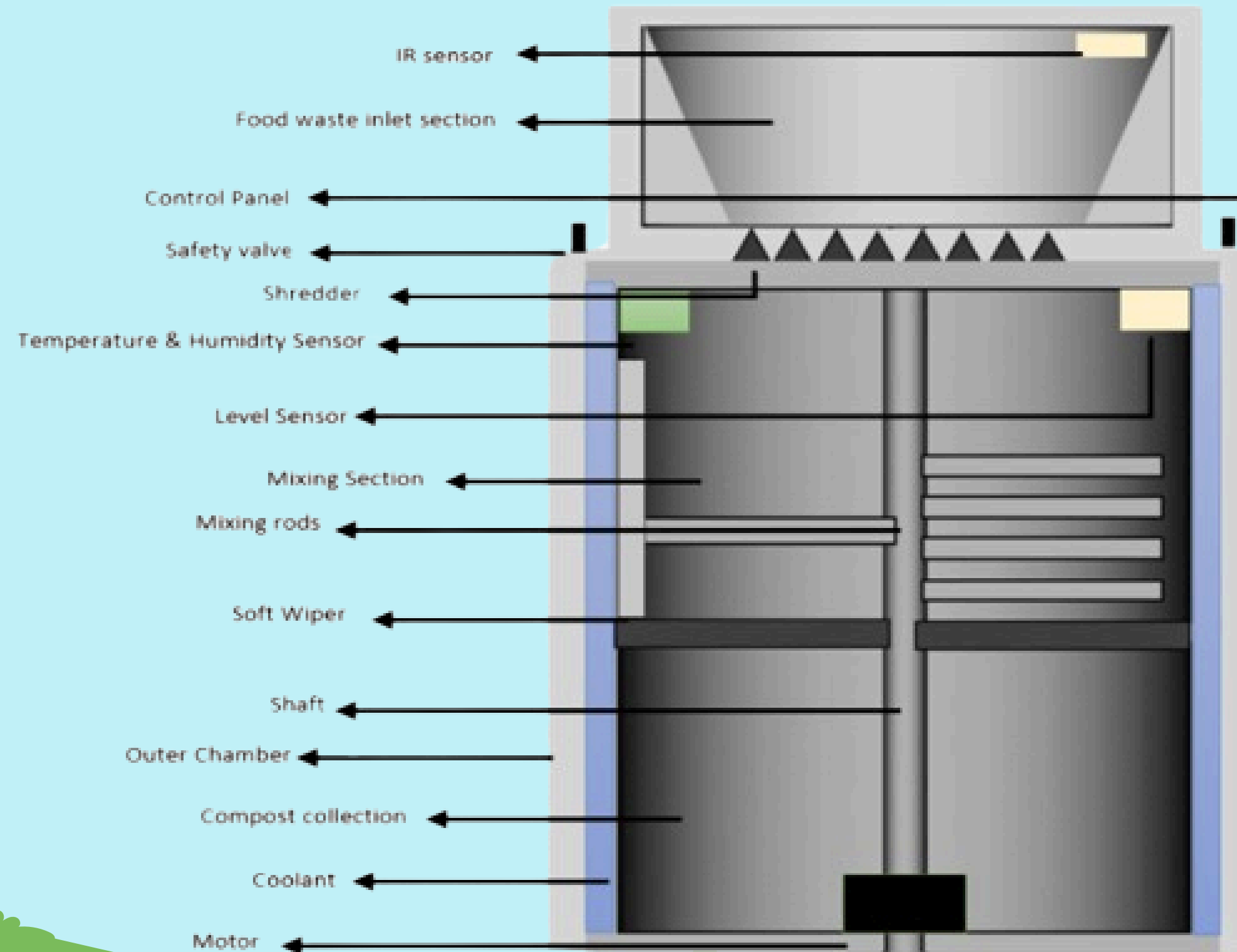


Lack of Proper waste management techniques



Solution

- The "e-Setch Bin" is a 24-hour compost bin that integrates advanced technology to efficiently convert food waste into compost
- Incorporating infrared sensors, shredders, level sensors, and temperature/humidity sensors to enhance composting processes.
- Automated features detect waste, optimize the composting process by drying the particles, regulate mixing for efficient composting ready in 24 hours.



Uniqueness

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Unlike traditional methods like the NADEP method or the Indore method, the **e-Setch Bin** offers a much quicker composting process

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Utilising **unique sensors** for automation

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e-Setch Bin **optimizes the composting process** to retain all nutrients within the compost unlike the Indore method

Revenue Model

Product Sales

- Selling compost
- Agricultural Lands
- Community Farms

Service Plans

maintenance
subscriptions with
app support

Value-added Products

Providing other
products such as
Biodegradable Bags
etc..



Scalability

1. Customised design for easy expansion.
2. Marketing through community farms
3. Robust distribution network for market penetration.



Impact

The e-Setch Bin reduces composting time from months to just 24 hours, significantly accelerating the process. This translates to a 90% reduction in composting time compared to traditional methods,

Enhances soil health and fertility, benefiting farm productivity.



Thank You!
Keep recycling!

